

NAVTTTC-NETSOL AIML Entry Test — Prep (Matched to Real Paper Format)

Based on the actual paper your friend solved, this test is a **Microsoft Forms quiz**, ~25 MCQs, 4 marks each, mixing:

1. Logical deduction / reasoning
2. Basic word-problem math (averages, discounts, rates, probability)
3. EQ-style workplace scenarios
4. Very basic AI/ML awareness (not deep theory)
5. Basic Python syntax

It is noticeably easier than a typical "ML interview" test — think **general aptitude + intro-level tech awareness**, not algorithms or math derivations. Good news: this plays to your strengths since you don't need to cram deep ML theory tonight.

1. Logical Deduction (Syllogisms)

These give you 2 premises and ask what can be validly deduced. Key rule: **only pick what MUST be true, not what's likely true.**

1. All programmers know coding. Ali is a programmer. What can you deduce? → **Ali knows coding.** (direct application of the premise — not "everyone who knows coding is a programmer," that reverses the logic)
2. All cats are animals. Some animals are black. Can we conclude all cats are black? → **No / Cannot be false as stated** — actually the valid answer is "**No**" — you cannot conclude this. (Some animals being black doesn't mean those animals are cats.)
3. All doctors are educated. Sara is educated. Is Sara a doctor? → **Cannot be determined** — being educated doesn't mean she's a doctor (reversed logic trap, same pattern as Q2 above).
4. All birds can fly. Penguins are birds. Can penguins fly? → Tests whether you follow the *given* premise even if factually wrong in real life — logic tests want you to accept premises as true and follow them. If premise says "all birds fly" then logically the deduction is "penguins can fly" **within the logic of the puzzle**, even though it's factually false. (Read the question style carefully — some tests want strict logical deduction, ignoring real-world knowledge.)
5. Some fruits are sweet. All apples are fruits. Are all apples sweet? → **Cannot be determined** — "some fruits" doesn't cover all fruits.

Pattern to remember: Watch for "all/some" mismatches — these are the classic trap. "All A are B" + "C is A" → "C is B" is valid. But "Some A are B" + "C is A" → does NOT mean "C is B."

2. Word Problems (Practice More of These — High-Value Section)

6. A farmer has 40 chickens. All but 12 are sold. How many does he have left? → **12** (this phrase means "all except 12 are sold" = 12 remain)
7. If 6 printers print 6 books in 6 minutes, how many printers are needed to print 60 books in 60 minutes? → **6** (rate is 1 printer = 1 book per 6 minutes; in 60 minutes one printer makes 10 books, so for 60 books you need 6 printers)
8. There are 30 students. Half leave, then 5 more leave. How many remain? → $30 \div 2 = 15$, then $15 - 5 = \mathbf{10}$
9. Calculate: $(18 \times 2) + (16 \div 4) - 5 \rightarrow 36 + 4 - 5 = \mathbf{35}$
10. A monitor costs \$400. A 25% discount is offered. What's the sale price? → $400 \times 0.75 = \mathbf{\$300}$
11. A shirt costs \$72 after a 10% discount. What was the original price? → $72 \div 0.9 = \mathbf{\$80}$
12. A number is multiplied by 4. If the original number is 7, what's the result? → **28**
13. Dataset: 5, 10, 15, 20, 25. What is the average? → $(5+10+15+20+25)/5 = 75/5 = \mathbf{15}$
14. A fair coin is tossed twice. Probability of exactly one head? → Outcomes: HH, HT, TH, TT → $2/4 = \mathbf{1/2}$
15. Euclidean distance between (2,3) and (5,7)? → $\sqrt{(5-2)^2 + (7-3)^2} = \sqrt{(9+16)} = \sqrt{25} = \mathbf{5}$

More practice (same style, try these yourself):

16. A shopkeeper buys an item for \$150 and sells it for \$180. What is the profit percentage? → Profit = \$30; % = $30/150 \times 100 = \mathbf{20\%}$
 17. If a car travels 150 km in 3 hours, what is its average speed? → **50 km/h**
 18. A class has 25 students. 40% are girls. How many boys are there? → Girls = 10, Boys = **15**
 19. What is $3/4$ of 200? → **150**
 20. Find the missing number: 2, 4, 8, 16, ? → **32** ($\times 2$ pattern)
 21. A book costs \$60. After two successive discounts of 10% each, what's the final price? → $60 \times 0.9 \times 0.9 = \mathbf{\$48.60}$
 22. If 5 workers complete a task in 10 days, how many days will 10 workers take (same rate)? → **5 days**
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3. EQ / Workplace Scenario Questions

Rule of thumb confirmed by the real paper: **pick the option that is calm, constructive, and communicative** — never the aggressive, avoidant, or passive-aggressive one.

23. A teammate is struggling to complete their work. Best response? → **Offer help if possible and discuss priorities**

- 24. Someone interrupts you repeatedly during a discussion. Best response? → **Wait for an appropriate moment and continue respectfully**
- 25. Your manager gives constructive feedback. Best response? → **Accept it positively and improve**
- 26. A colleague asks for help on something you understand well. Best response? → **Help them if your workload allows**
- 27. A misunderstanding occurs between two teammates. Best approach? → **Encourage respectful communication to resolve it**

More practice:

- 28. You disagree with a decision made by your team lead. Best response? → Respectfully share your perspective and reasoning, then support the team's final decision.
 - 29. You made a mistake that affected a colleague's work. Best response? → Acknowledge it, apologize, and offer to help fix it.
 - 30. A team member takes credit for your idea in a meeting. Best response? → Calmly clarify your contribution afterward in a professional, non-confrontational way.
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4. Basic AI/ML Awareness (Not Deep Theory)

- 31. Which of the following is an application of Machine Learning? → **Email spam detection**
- 32. Artificial Intelligence is mainly used to: → **Make machines perform intelligent tasks**
- 33. Which of the following is an example of supervised learning? → **Classification**

More practice (same easy level):

- 34. Which of these is an example of unsupervised learning? → Clustering (e.g., grouping customers by behavior — no labeled output)
 - 35. What does "training a model" mean in simple terms? → Teaching a program to recognize patterns using example data
 - 36. Which of these is NOT typically an AI application? → Basic calculator arithmetic (rule-based, not learned)
 - 37. Recommendation systems (like Netflix suggesting movies) are an example of: → Machine Learning application
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5. Basic Python Syntax

- 38. Which function is used to display output in Python? → **print()**
- 39. Which symbol is used to assign a value to a variable in Python? → **=**

40. Which data type stores whole numbers? → **int**
41. What does `len("Python")` return? → **6**
42. Which keyword is used to define a function in Python? → **def**

More practice:

43. Which symbol is used for comments in Python? → **#**
44. What is the output of `print(2 ** 3)`? → **8**
45. Which of these is a valid Python list? → `[1, 2, 3]`
46. What does `type(5.0)` return? → `<class 'float'>`
47. Which loop is used to iterate a fixed number of times in Python? → `for` loop
48. What does `input()` do in Python? → Takes user input from the keyboard (returns it as a string)
49. How do you check if a variable `x` equals 5 in Python? → `x == 5` (double equals for comparison, single `=` is assignment)
50. What will `"5" + "5"` return in Python? → **"55"** (string concatenation, not addition, since both are strings)
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Test-Day Strategy (Updated)

- This test rewards **speed and accuracy on simple things** more than depth — don't overthink the math or logic questions.
- For "all/some" logic questions, slow down for 5 extra seconds — that's the main trap type on this paper.
- Word problems: read carefully for phrases like "all but," "half then more leave," "successive discounts" — the wording is the trick, not the math itself.
- EQ questions: always the calm/constructive/communicative option.
- Python questions are genuinely basic (`print`, `=`, `int`, `len`, `def`) — you already know all of this cold from your coursework.
- No calculator needed — everything is designed for quick mental math.

You're in a strong position for this one — your CS background covers this comfortably. Focus your remaining prep time on speed drills for the word problems and logic traps rather than the AI/ML or Python sections.